USN | | | | | | | | | | [23ESME013]

FIRST SEMESTER B. TECH EXAMINATION, JANUARY - 2025

BASICS OF CIVIL AND MECHANICAL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART - A			**5x2=10**
1.	**a.**	List the types of power plants.	1	1	**02**
	b.	Define welding, soldering and brazing.	1	2	**02**
	c.	Differentiate between spark ignition and compression ignition engines.	2	3	**02**
	d.	Enumerate the importance of civil Engineering in development of a country.	2	4	**02**
	e.	What is bridge and list the components of the bridges.	2	5	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain in detail the working principle of two stroke petrol engines.	2	3	**05**
		OR			
	b.	List the types of nuclear reactors.	2	1	**05**
3.	**a.**	Compare steam, hydro and nuclear powerplants.	2	1	**05**
		OR			
	b.	Explain in detail the TIG and MIG welding.	2	2	**05**
4.	**a.**	Explain the fins, evaporators, condensers and radiators.	2	2	**05**
		OR			
	b.	Define **i.** Compression ratio. **ii.** Top dead center **iii.** Cylinder volume **iv.** Bore **v.** Swept volume	1	3	**05**
5.	**a.**	Enumerate on cylinder lines in IC engines.	2	3	**05**
		OR			
	b.	List the components of gas turbine powerplant and describe its functions.	1	1	**05**
6.	**a.**	What are composite materials? Classify the composite materials.	2	3	**05**
		OR			

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	b.	Discuss the emerging areas in civil engineering.	2	4	**05**
7.	**a.**	Explain in detail about water security.	2	4	**05**
		OR			
	b.	Explain the importance of rain water harvesting.	2	3	**05**
		PART - C		**6x10=60**	
8.	**a.**	Explain in detail the diesel powerplant.	2	1	**10**
		OR			
	b.	Give the layout of a simple gas turbine plant and explain its working. Also bring out the advantages and disadvantages of using gas turbine power plant for power generation.	2	1	**10**
9.	**a.**	Briefly explain metal matrix and ceramic matrix composite.	2	2	**10**
		OR			
	b.	Enumerate on properties of metals and non-metals.	2	2	**10**
10.	**a.**	Explain internal combustion engines components.	2	3	**10**
		OR			
	b.	Explain in detail the 2 stroke and 4 stroke diesel engines.	2	3	**10**
11.	**a.**	What is hydel energy? With a neat layout of a hydel power plant, describe it's working. What are the advantages and disadvantages of power generation?	2	1	**10**
		OR			
	b.	What are the main circuits of steam power plant?	1	1	**10**
12.	**a.**	Explain the global warming and construction industry.	2	5	**10**
		OR			
	b.	Explain in detail the environmental issues concerned to building materials.	2	5	**10**
13.	**a.**	Explain the advantages and disadvantages of automation and robotics in construction.	2	4	**10**
		OR			
	b.	Explain in detail civil aspects of pipe design-sewer design.	2	5	**10**